

# XIANGLIN HUANG

## CURRICULUM VITAE

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**Position:** Visiting Research Professor, Dongguk University    [ORCID](#)    [impacthuang.github.io](https://github.com/impacthuang)

**Overview:** I am a postdoctoral researcher at KAIST and a Visiting Research Professor at Dongguk University, specializing in the impact safety and prognostics & health management (PHM) of lithium-ion batteries, and in data-driven constitutive modeling of metals under impact loading. At the Dongguk iPHM Lab, I lead an independent research stream on lithium-ion battery impact safety, co-supervising graduate students on experimental failure-mode characterization, deep-learning frameworks for acoustic-emission-based damage detection, and multiphysics finite-element simulation of battery packs under dynamic loading.

In parallel, I specialize in data-driven constitutive modeling of metals under impact loading. A four-paper series I authored in the International Journal of Impact Engineering (2023–2027) establishes a complete framework, combining SVD/CP tensor decomposition with artificial neural networks, for verifying, determining, and validly extrapolating dynamic flow stress equations from discrete SHPB and Taylor-Hopkinson experimental data.

I have authored 17 SCIE-indexed journal papers (16 in Q1 journals), 6 as first author, and currently serve as a graduate course instructor at Dongguk University and peer reviewer for seven international journals, including the International Journal of Impact Engineering, Thin-Walled Structures, and Journal of Mechanical Science and Technology.

### Summary

|                             |     |                                   |    |
|-----------------------------|-----|-----------------------------------|----|
| Total Journal Publications  | 17  | Q1 Journal Publications           | 16 |
| First-Authored Publications | 6   | Corresponding-Author Publications | 5  |
| Conference Presentations    | 4   | Peer Review Roles                 | 7  |
| Total Citations             | 369 | h-index                           | 8  |

### Education

Aug 2019 –  
Dec 2024

**Ph.D. in Mechanical Engineering**  
University of Manchester, Manchester, UK  
Department of Mechanical, Aerospace and Civil Engineering, School of Engineering  
Thesis: *Determination of dynamic flow stress equation based on discrete material data*  
Supervisor: Prof. Qingming Li (Q.M. Li), [Qingming.li@manchester.ac.uk](mailto:Qingming.li@manchester.ac.uk)

Sep 2015 –  
Jul 2017

**M.E. in Aeronautical and Astronautical Science and Technology**  
Harbin Institute of Technology (HIT), Harbin, China  
School of Astronautics  
Thesis: *Experimental investigation on the impact resistance of polymer-aluminium layered plates*  
Supervisor: Prof. Wei Zhang, [zhdawei@hit.edu.cn](mailto:zhdawei@hit.edu.cn)

Sep 2010 –  
Jul 2014

**B.E. in Aircraft Manufacturing Engineering**  
Civil Aviation University of China (CAUC), Tianjin, China  
College of Aeronautical Engineering  
Coursework: Mechanics of Materials, Aircraft Fault Diagnosis Technology, Engineering Materials, Aircraft Flight Mechanics, Aircraft Manufacturing Technology

# Academic & Professional Appointments

|                     |                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2025 – Present  | <p><b>Visiting Research Professor</b></p> <p>Dept. of Mechanical, Robotics and Energy Engineering, Dongguk University–Seoul</p> <p>Supervisor: Prof. Heung Soo Kim</p> <ul style="list-style-type: none"><li>• Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework</li></ul> |
| Oct 2025 – Present  | <p><b>Postdoctoral Researcher</b></p> <p>InnoCORE PRISM-AI Center, KAIST, South Korea</p> <p>Supervisor: Prof. Seunghwa Ryu</p> <ul style="list-style-type: none"><li>• Prognostics and Health Management of Lithium-Ion Batteries Based on Neuromorphic Sensor and Artificial Intelligence Framework</li></ul>                                        |
| May 2025 – Aug 2025 | <p><b>Research Associate (Part-time)</b></p> <p>Zhejiang University, China</p> <p>Supervisor: Prof. Guoxing Lu</p>                                                                                                                                                                                                                                     |
| Oct 2020 – Apr 2024 | <p><b>Research Associate</b></p> <p>University of Manchester, UK</p> <p>Supervisor: Prof. Q.M. Li</p>                                                                                                                                                                                                                                                  |

# Teaching Experience

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar 2026 – Jun 2026 | <p><b>Instructor (Graduate Course)</b></p> <p>Dongguk University, “Machine Learning and Its Engineering Application”</p> <ul style="list-style-type: none"><li>• Sole instructor for a graduate-level course covering CNN, RNN, LSTM, and Transformer architectures, with emphasis on their application to mechanical and PHM systems; students implemented models hands-on for data-driven diagnosis and prediction problems.</li><li>• Mentored a team of students competing in the 2nd KSPHM-KIMM Mechanical Data Challenge 2026, a national mechanical data-driven diagnostics competition (concluded June 25, 2026).</li></ul> |
| Sep 2019 – Dec 2023 | <p><b>Graduate Teaching Assistant</b></p> <p>University of Manchester, UK</p> <ul style="list-style-type: none"><li>• Supported students completing the course Advanced Modelling &amp; Simulation</li><li>• Supported students completing the course Composites &amp; Polymers</li><li>• Demonstrated the operating procedure of column buckling experiments</li><li>• Instructed students completing ANSYS numerical simulation coursework</li></ul>                                                                                                                                                                              |

# Research Experience

My research spans two connected streams built on a common foundation of experimental impact mechanics: the impact safety and prognostics & health management (PHM) of lithium-ion batteries, and a data-driven framework for the dynamic constitutive modeling of metals.

2023 –  
Present

### **Lithium-Ion Battery Impact Safety and Prognostics & Health Management (PHM)**

- Leading a battery PHM research group at the Dongguk iPHM Lab, developing neuromorphic-sensor- and AI-based frameworks for lithium-ion battery health monitoring.
- Co-supervised, as corresponding author, experimental characterization of lithium-ion battery failure modes and failure-energy thresholds under extreme impact, and multiphysics finite-element characterization of critical failure impact on battery pack stability. *Journal of Energy Storage* 150 (2026) 120037; *Engineering Failure Analysis* 196 (2026) 111084.
- Co-supervised, as corresponding author, a deep-learning framework using acoustic emission signals for real-time damage assessment of lithium-ion batteries under dynamic impact (CAUC-funded Graduate Research Project). *Journal of Power Sources* 662 (2026) 238737.
- Authored an invited review of the battery PHM field, synthesizing current challenges and future perspectives. *Journal of Mechanical Science and Technology* 40(4) (2026) 2405–2415.
- Drafted the rationale, research content, and literature review for a National Natural Science Foundation of China (NSFC) grant proposal on multiscale, multiphysics-coupled failure mechanisms of high-energy-density batteries under dynamic loading.

2019 –  
2027

### **Data-Driven Constitutive Modeling of Dynamic Flow Stress**

- Verified when decoupled, Johnson-Cook-type flow stress equations are mathematically legitimate for a given material, using the rank-1 energy ratio of an SVD decomposition of two-way flow stress data as a quantitative criterion. *International Journal of Impact Engineering* 173 (2023) 104453.
- Determined dynamic flow stress equations directly from discrete, non-ideal SHPB data by relaxing the constant-strain-rate requirement of classical reduction protocols, resolving the strain-rate dependence (Part 1) and extending to the full three-dimensional strain, strain-rate and temperature surface via CP tensor decomposition (Part 2). *International Journal of Impact Engineering* 206 (2025) 105403 and 105432.
- Developed TEDI-FS (Tensor-Decomposition-based Extrapolation and Dynamic Identification of Flow Stress), a framework that validly extrapolates an SHPB-calibrated flow stress model to very-high strain rates ( $\sim 10^5 \text{ s}^{-1}$ ), inversely identified and validated against coupled Taylor-Hopkinson impact experiments and explicit finite element simulation. *International Journal of Impact Engineering* 219 (2027) 105863.
- The methodology combines SVD/CP tensor decomposition with artificial neural networks and applies beyond the Johnson-Cook equation to any factorized constitutive model, built on an explicit isotropic-metal (J2 plasticity) assumption stated in the underlying Ph.D. thesis.

2015 –  
2026

### **Impact Behavior of Composite and Sandwich Structures**

- Co-supervised, as corresponding author, ANN surrogate models for the ballistic resistance and residual-velocity prediction of finite-thickness metallic plates under impact. *Thin-Walled Structures* 203 (2024) 112161; *Thin-Walled Structures* 216 Part B (2025) 113685.
- Contributed constitutive modeling and simulation analysis to the multiscale modeling of impact behavior and mesoscale damage in 3D angle-interlock woven composites. *International Journal of Mechanical Sciences* 276 (2024) 109382; *Composite Structures* 382 (2026) 120111.
- During master's research at HIT, designed and executed experiments, including DIC-based penetration testing, and finite element simulations for the dynamic response of sandwich structures and polymer-aluminum laminated plates under impulsive and ballistic loading; first author on the resulting *International Journal of Impact Engineering* 113 (2018) paper, co-author on three further *International Journal of Impact Engineering* papers (2016 to 2017), and one AIP Conference Proceedings paper.

## **Student Supervision**

*Master's students, co-supervised with Dr. Yunfei Deng:*

2022 – 2025

**Xiaoyue Yang**

- Impact mechanics; ANN surrogate models for the impact resistance of finite-thickness plates.
- Co-authored: *Thin-Walled Structures* 203 (2024) 112161; *Thin-Walled Structures* 216 Part B (2025) 113685.

→ Now: Ph.D. Candidate, Hunan University

2023 – 2026

**Han Zheng**

- Mechanical safety of lithium-ion batteries: failure modes under impact and multiphysics response at the pack level.
- Co-authored: *Journal of Energy Storage* 150 (2026) 120037; *Engineering Failure Analysis* 196 (2026) 111084.

→ Now: Ph.D. Student, Sun Yat-sen University

2023 – 2026

**Jiangtao Li**

- Deep learning for real-time damage assessment of lithium-ion batteries under dynamic impact, with acoustic emission monitoring.
- Co-authored: *Journal of Power Sources* 662 (2026) 238737.

→ Now: Ph.D. Candidate, Beijing Institute of Technology

2023 – 2026

**Yixu Lv**

- Neural network surrogate models for residual velocity prediction in finite-thickness plates.
- Co-authored: *Thin-Walled Structures* 216 Part B (2025) 113685.

→ Now: Materials Engineer, AEROFUGIA

***Undergraduate & postgraduate thesis supervision, University of Manchester:***

- Instructed undergraduate student thesis: Machine learning based dynamic constitutive models (10/2023 – 04/2024)
- Instructed postgraduate (master's degree) student thesis: Signal analysis of damaged space structure by impact (08/2020 – 07/2021)
- Instructed undergraduate student thesis: Structural health monitoring of space structures under impacts (10/2020 – 03/2021)

## Awards & Grants

May 2022

**UK Henry Lester Trust Grant**

University of Manchester, UK

May 2021

**UK Henry Lester Trust Grant**

University of Manchester, UK

## Journal Publications (SCIE-indexed: 17, Q1: 16)

### 2027

1. **X. Huang**, Q.M. Li\*, Tensor-decomposition-based inverse characterization of flow stress for very-high strain-rate impact, *International Journal of Impact Engineering*, 219 (2027) 105863. IF: 6.7, Q1

### 2026

1. Y. Deng, H. Zheng, T. Zhang\*, **X. Huang\***, Characterizing critical failure impact on LIB pack stability via multiphysics response, *Engineering Failure Analysis*, 196 (2026) 111084. IF: 7.0, Q1
2. **X. Huang**, H.S. Kim\*, Advancing battery prognostics and health management: Challenges and future perspectives, *Journal of Mechanical Science and Technology*, 40(4) (2026) 2405–2415. IF: 2.2, Q3
3. H. Wei, **X. Huang**, X. Jiang, W. Xie, Y. Li, G. Zhao, W. Zhang\*, Mesoscale damage modeling of 3D woven composites under quasi-static loads: Incorporating plasticity and fiber misalignment, *Composite Structures*, 382 (2026) 120111. IF: 7.8, Q1
4. Y. Deng, H. Zheng, **X. Huang\***, Failure modes and failure energy threshold of lithium-ion batteries under extreme impact, *Journal of Energy Storage*, 150 (2026) 120037. IF: 10.7, Q1
5. Y. Deng, J. Li, **X. Huang\***, Deep learning-based real-time damage assessment of lithium-ion batteries under dynamic impact, *Journal of Power Sources*, 662 (2026) 238737. IF: 8.4, Q1

## **2025**

1. **X. Huang**, Q.M. Li\*, Determination of dynamic flow stress equation based on discrete experimental data: Part 1 Methodology and the dependence of dynamic flow stress on strain-rate, *International Journal of Impact Engineering*, 206 (2025) 105403. IF: 6.7, Q1
2. **X. Huang**, Q.M. Li\*, Determination of dynamic flow stress equation based on discrete experimental data: Part 2 Dynamic flow stress depending on strain, strain-rate and temperature, *International Journal of Impact Engineering*, 206 (2025) 105432. IF: 6.7, Q1
3. Y. Deng, Y. Lv, X. Yang, C. Du, **X. Huang\***, Determination of residual velocity model of finite thickness material based on Artificial Neural Network, *Thin-Walled Structures*, 216 Part B (2025) 113685. IF: 8.3, Q1

## **2024**

1. H. Wei, **X. Huang**, W. Xie, X. Jiang, G. Zhao, W. Zhang\*, Multiscale modeling for the impact behavior of 3D angle-interlock woven composites, *International Journal of Mechanical Sciences*, 276 (2024) 109382. IF: 11.4, Q1
2. Y. Deng, X. Yang, **X. Huang\***, Determination of ballistic resistance model of finite thickness material based on Artificial Neural Network, *Thin-Walled Structures*, 203 (2024) 112161. IF: 8.3, Q1

## **2023**

1. **X. Huang**, Q.M. Li\*, The legitimacy of decoupled dynamic flow stress equations and their representation based on discrete experimental data, *International Journal of Impact Engineering*, 173 (2023) 104453. IF: 6.7, Q1

## **2022**

1. Y. Deng, H. Wu, Y. Zhang\*, **X. Huang**, X. Xiao, Y. Lv, Experimental and numerical study on the ballistic resistance of 6061-T651 aluminum alloy thin plates struck by different nose shapes of projectiles, *International Journal of Impact Engineering*, 160 (2022) 104083. IF: 6.7, Q1

## **2018**

1. **X. Huang**, W. Zhang\*, Y. Deng, X. Jiang, Experimental investigation on the ballistic resistance of polymer-aluminum laminated plates, *International Journal of Impact Engineering*, 113 (2018) 212–221. IF: 6.7, Q1

## **2016–2017**

1. N. Ye, W. Zhang\*, D. Li, W. Huang, W. Xie, **X. Huang**, X. Jiang, Dynamic response and failure of sandwich plates with PVC foam core subjected to impulsive loading, *International Journal of Impact Engineering*, 109 (2017) 121–130. IF: 6.7, Q1
2. W. Huang, W. Zhang\*, **X. Huang**, X. Jiang, Y. Li, L. Zhang, Dynamic response of aluminum corrugated sandwich subjected to underwater impulsive loading: Experiment and numerical modeling, *International Journal of Impact Engineering*, 109 (2017) 78–91. IF: 6.7, Q1
3. W. Huang\*, B. Jia, W. Zhang\*, **X. Huang**, D. Li, P. Ren, Dynamic failure of clamped metallic circular plates subjected to underwater impulsive loads, *International Journal of Impact Engineering*, 94 (2016) 96–108. IF: 6.7, Q1

## **Conferences**

### **Presentations**

1. **X. Huang** (Invited Speaker), H.S. Kim\*, “Impact-Induced Damage Assessment and Remaining Useful Life Prediction of Lithium-Ion Cells Using Acoustic Emission Features,” *PHM Korea 2026 Annual Conference*, Busan, Republic of Korea, Jun 2026
2. **X. Huang** (Oral Presentation), H.S. Kim\*, “Acoustic Emission Characteristics of Lithium-Ion Cells under Mechanical Impact,” *2026 Spring Conference of the Korean Society for Noise and Vibration Engineering (KSNVE)*, Gangwon-do, Republic of Korea, May 2026
3. **X. Huang** (Oral Presentation), H.S. Kim\*, “Impact-Induced Pulse Power Capability Degradation in Lithium-Ion Cells: Experimental Characterization and Structural Response Analysis,” *2026 Spring Conference of the CAE and Applied Mechanics Division, KSME*, Yeosu-si, Republic of Korea, May 2026
4. **X. Huang** (Oral Presentation), H.S. Kim\*, “Physics-Informed Probabilistic Machine Learning for Battery Degradation Diagnostics under Dynamic Mechanical and Electrochemical Conditions,” *KSME 3rd Emerging Technologies in Mechanical Engineering (ETME 2025)*, Kangwon Land, Republic of Korea, Dec 2025

### **Published Proceedings**

1. W. Zhang\*, **X. Huang**, Y. Qi, D. Li, J. Tao, W. Huang, Experimental investigation on ballistic stability of high-speed projectile in sand, *AIP Conference Proceedings* 1793, 060031 (2017)

## **Technical Skills**

- **Simulation Software:** ABAQUS, LS-DYNA: explicit modelling of dynamic impact with user-defined subroutines (VUMAT/UMAT)
- **Programming Languages:** C, Python, MATLAB, Fortran
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **CAD / Geometric Modelling:** CATIA

## **Peer Review Board Membership (7 journals)**

- *International Journal of Impact Engineering* (Elsevier)
- *Scientific Reports* (Springer Nature)
- *Defence Technology* (Elsevier)
- *Measurement* (Elsevier)
- *Thin-Walled Structures* (Elsevier)
- *Journal of Mechanical Science and Technology* (Springer)
- *Journal of Advanced Energy Research*

## **References**

### **Prof. Qingming Li (Q.M. Li), Ph.D. Supervisor**

- Professor, School of Engineering, University of Manchester, UK
- Email: [Qingming.li@manchester.ac.uk](mailto:Qingming.li@manchester.ac.uk)

### **Prof. Heung Soo Kim, Current Supervisor**

- Director, BK21 AIMS Center; Director, DGU Global Smart Factory Center
- Dept. of Mechanical, Robotics and Energy Engineering, Dongguk University, South Korea
- Email: [heungsoo@dgu.edu](mailto:heungsoo@dgu.edu)

### **Prof. Seunghwa Ryu, Current Supervisor**

- InnoCORE PRISM-AI Center, Korea Advanced Institute of Science and Technology (KAIST), South Korea
- Email: [ryush@kaist.ac.kr](mailto:ryush@kaist.ac.kr)